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# What Frequency Matching Might (not) Be

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## Abstract

Frequency matching has been proposed as a central principle underlying phonological knowledge and learning. A central example comes from Hungarian vowel harmony, where stems with back vowels followed by neutral vowel(s) show structurally-conditioned lexical variation in whether they take back or front suffixes. In wug tests, Hungarian speakers show evidence of internalizing this structural conditioning, often taken as evidence of speakers internalizing a probabilistic grammar. This article critically analyzes the Frequency Matching Hypothesis and argues that the appearance of frequency matching emerges from variation across speaker vocabularies and from averaging over categories that may not reflect those internalized by speakers. I suggest that the central question of lexical variation is *how learners discover its structural conditioning* and propose a learning theory, in the form of an independently-motivated learning model, for that question. Assuming just variation across speaker vocabularies, the model exhibits the appearance of frequency matching. The model matches wug test results as well as previously proposed probabilistic grammars, predicts a diachronic shift in Hungarian vowel transparency, and explains a surprising lack of token-level variation in Hungarian vowel harmony.

## 1. Introduction

For at least the last two and a half decades, the concept of *frequency matching* has risen in prominence in phonology (Zuraw 2000; Albright & Hayes 2003; Ernestus & Baayen 2003; Hayes & Londe 2006; Hayes *et al.* 2009; Zuraw 2010; Becker *et al.* 2011; Gouskova & Becker 2013; Linzen *et al.* 2013; Zymet 2018; Kuo 2024; Liang *et al.* 2024; Jarosz *et al.* 2025, among others). The concept is commonly applied to cases of *lexical variation*, in which there is variation in phonological behavior across lexical items. While lexical variation is unpredictable in the strict sense that the behavior of particular items is idiosyncratic, the variation is almost always *structurally conditioned*: there are features of lexical items that allow for grouping them such that the aggregate behavior is somewhat consistent within groups. A central case comes from Hungarian vowel harmony. The basic generalization is backness harmony (Siptár & Törkenczy, 2000): stems ending in back (B) vowels take back suffixes and those ending

in front (F) vowels take front suffixes (1b). But the behavior is variable for stems with a back vowel followed by one or more front unround vowels, often called the *neutral* (N) vowels (1c). The vowel inventory is in (1a).

|             |    | Front           |                |       |     | Back |     |      |
|-------------|----|-----------------|----------------|-------|-----|------|-----|------|
|             |    | unround         |                | round |     |      |     |      |
| (1)         | a. | High            | [i]            | [i:]  | [y] | [y:] | [u] | [u:] |
|             |    | Mid             |                | [e:]  | [ø] | [ø:] | [o] | [o:] |
|             |    | Low             | [ɛ]            |       |     |      | [ɔ] | [ɑ:] |
|             | b. | [mo:kʉf-nək]    | “squirrel”-DAT |       |     |      |     |      |
|             |    | [mɔtʃkɔ-nək]    | “cat”-DAT      |       |     |      |     |      |
|             |    | [øɾəm-nək]      | “joy”-DAT      |       |     |      |     |      |
|             |    | [køɲv-nək]      | “book”-DAT     |       |     |      |     |      |
|             | c. | [kotʃi-nək]     | “car”-DAT      |       |     |      |     |      |
|             |    | [kontʃɛɾt-nək]  | “rope”-DAT     |       |     |      |     |      |
|             |    | [pɑ:tsiɛɲf-nək] | “patient”-DAT  |       |     |      |     |      |
| [dosie:nək] |    | “dossier”-DAT   |                |       |     |      |     |      |

Which suffix a stem takes is idiosyncratic, so this is lexical variation. Nevertheless, there are features that allow for predicting whether a stem is likely to be back-suffix-taking or front-suffix-taking. The strongest predictors are often described as **height**: stems with higher N vowels are more likely to take a back suffix (high [i, i:] » mid [e:] » low [ɛ]) and **count**: stems with back vowels followed by two neutral vowels (BNN stems) are more likely to take the front suffix than those where only one N follows (BN stems). Thus, the lexical variation is structurally conditioned. Hayes & Londe (2006, henceforth H&L) investigated the categories formed by taking the Cartesian product (or interaction) of these two variables (2).

|     |                     |                    |                    |
|-----|---------------------|--------------------|--------------------|
| (2) | BN <sub>high</sub>  | BN <sub>mid</sub>  | BN <sub>low</sub>  |
|     | BNN <sub>high</sub> | BNN <sub>mid</sub> | BNN <sub>low</sub> |

H&L constructed a corpus of Hungarian datives and computed, within each of these categories of interest, the fraction of stems that took back suffixes. The height and count effects, together with their interaction, are shown in Figure 1 for H&L’s dataset (see § 4.1.2 for dataset description), where the y-axis shows the fraction of words that take back suffixes. H&L created wugs (Berko, 1958) for each category and asked Hungarian speakers to decide whether each wug takes a back or a front suffix. Aggregating

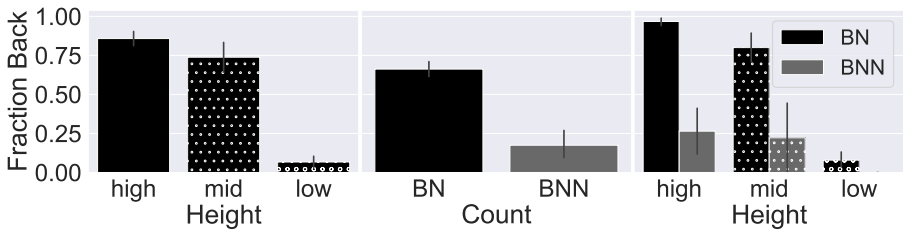


Figure 1: Some previously-reported structural conditioning effects.

responses within each category, H&L found that the fraction of back suffix responses in each category roughly approximated the lexical frequencies. In their view, this showed that speakers were frequency matching.

There are three propositions that I believe are central to what I will call the *frequency matching hypothesis*.

### (3) The Frequency Matching Hypothesis

- a. Lexical variation is structurally conditioned.
- b. Learners' productive responses in wug tests, in the aggregate, largely mirror the structurally-conditioned lexical frequencies.
- c. The observed similarity between the structural conditioning of the lexicon (input) and productive output is a consequence of individuals internalizing a probabilistic grammar.

The first two are empirical observations while the third is a proposed mechanism or causal explanation. The first, (3a), is uncontroversial. The second, (3b), has broad empirical support (e.g., Zuraw 2000; Ernestus & Baayen 2003; Hayes & Londe 2006; Hayes *et al.* 2009; Gouskova & Becker 2013), though apparent deviations are met with interest (Goldrick & Larson, 2010; Becker *et al.*, 2011; Liang *et al.*, 2024). These deviations are not typically taken as evidence against frequency matching, but rather of Universal Grammar biasing learners away from veridical distributional learning. This negative view of UG suppressing learning is an inversion of the traditional view of UG as the inherited formal machinery that makes learning language possible (Chomsky, 1955, 1965). Regardless, the basic view (3a)-(3b) convey is that learners' collective productive output is, putting aside some interesting discrepancies, similar in statistical shape to the input. Because the input to one is an aggregation of output from others, this is perhaps not surprising. But, since the output under consideration is productive behavior over wugs, the claim is that lexical variation is represented in some way in the grammar, not just marked idiosyncratically.

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While the first two propositions are probably universal to views on frequency matching in lexical variation, the inclusion of (3c) requires some justification. Hayes *et al.* (2009, p.826) describe the “Law of Frequency Matching” as the finding that “Speakers of languages with variable lexical patterns respond stochastically when tested on such patterns. Their responses aggregately match the lexical frequencies.” This clearly includes (3a)-(3b) but only claims that an individual’s responses are stochastic and that similar statistical shapes arise from *aggregating* over individuals. But the authors’ position appears to include (3c). H&L say it clearly, “we find that native speakers are good frequency matchers” (p.61), that “when confronted with a novel wug stem, each speaker behaved stochastically, in a way that matched the frequencies of the lexicon” (p.74; note “each speaker” and “stochastically”), and that “it is evident from our experimental results that Hungarian-learning children ultimately develop a STATISTICAL MODEL of the lexicon” (p.91). Similarly, Hayes *et al.* (2009, henceforth HZSL) describe H&L’s results as demonstrating that the statistical patterns in the lexicon are “internalized by native speakers, following the law of frequency matching” (p.833) and Ernestus & Baayen (2003)’s as showing “that Dutch speakers do tacitly know the quantitative alternation pattern of the lexicon” (p.826). They take their own results to indicate “that at a basic level, the participants were following the law of frequency matching, projecting the patterns of the lexicon into their own wug-test intuitions” (p.849). It is difficult to read these as anything other than a claim about individuals. Moreover, the authors proceed to analyze the aggregate responses via probabilities assigned by a MaxEnt grammar. Assuming they believe grammars to be properties of individuals not populations, they are assuming that population-level variation (approximated via the wug test) is represented by a probabilistic, population-level grammar that is simultaneously internalized by each individual. This is (3c).

To separate the first two propositions from the third, I call (3a)-(3b) the *Lexical Variation in the Grammar Hypothesis*. This hypothesis agrees that lexical variation, rather than being marked idiosyncratically, enters into the grammar. But the Frequency Matching Hypothesis goes further, postulating that the lexical variation is expressed via stochasticity in each learner’s grammar.

It is worth distinguishing the (3) sense of *lexical frequency matching* from the variational sociolinguistic sense of *probability matching*. These differ both in whether they come with a commitment to a probabilistic mechanism (the latter does not) and in being centrally concerned with type-level vs. token-level variation. The Frequency Matching Hypothesis takes probability matching to be not just a description of facts (3a-3b) but also to imply a central probabilistic mechanism (3c) underlying lexical variation,

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where the variation is type-level (concentrated across lexical items). In contrast, probability matching in variational sociolinguistics typically refers to token-level variation (probabilities of variants of an item) and is taken to be an empirical observation with no necessary commitment to any particular underlying mechanism: “it is not a hypothesis that children do probability matching. It is simply a description of the observed facts” (Labov, 1994, p. 583). The critique of frequency matching in this paper is centered on the former—the claim that lexical variation centrally involves individual learners matching type-level frequencies learned and represented by probabilistic mechanisms.

In what follows, I critically analyze The Frequency Matching Hypothesis, focusing on H&L and HZSL’s now foundational works on frequency matching. I argue that while these works support the Lexical Variation in the Grammar Hypothesis, they fail to demonstrate the Frequency Matching Hypothesis. More strongly, the study designs preclude the possibility of them doing so (§ 2). However, this critical analysis is not an argument against the claim that wug test productions mirror the lexical distribution. Nor is it an argument that learners never probability match (Labov, 1989; Roberts, 1997; Smith *et al.*, 2009). Instead, I propose a constructive way forward. I argue that the central mechanistic question of lexical variation is a learning question: *how do learners discover the structural conditioning of lexical variation?* Without an independently-motivated and independently-evaluated approach to this question, it is not possible to determine whether individuals match type-level frequencies. I propose a psychologically-motivated, learning-based approach to the question of structural conditioning (§ 3), and demonstrate how the approach allows for lexical variation to be more mechanistically understood. The proposed learning model accounts for H&L and HZSL’s wug test data at least as well as those authors’ probabilistic grammar proposal (§ 4). The appearance of frequency matching emerges from averaging over categories that may not reflect those internalized by speakers and from ordinary variation across learner vocabularies. I then present evidence directly for this proposal (§ 5). I demonstrate that the limited amount of token-level variation (the *vacillators*) in Hungarian vowel harmony follows from this learning theory. Moreover, the theory predicts that some but not all *transparent* vowels in Hungarian may be subject to opacification over time and that these and only these vowels indeed show evidence of opacification over time. I conclude in (§ 6).

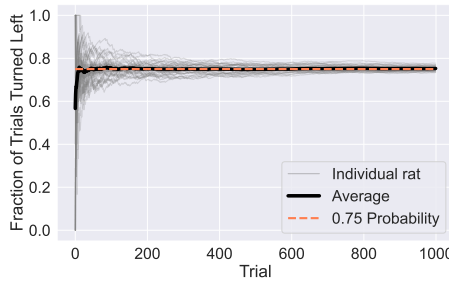


Figure 2: Simulated, hypothetical rats running a probabilistic T-maze using a deterministic, non-rate-tracking strategy. The rats still probability match.

## 2. Critical Analysis

### 2.1. The General Problem of the Causal Structure of Frequency

Frequency effects abound, but we must be cautious in drawing inferences about underlying causal structure (Caplan & Durvasula, 2026). Indeed, probability matching behavior is observed across the animal kingdom (Brunswik, 1939; Estes, 1964; Bitterman, 1965; Gallistel, 1990). For example, in making a choice (e.g., to turn left or right in a T-maze), if one choice (e.g., left) is rewarded with 0.75 probability and the other with 0.25 probability, various animals will, under some experimental conditions, match the probabilities, turning left 75% of the time and right 25%.

It is tempting to assume that this behavior is due to animals tracking the rates of outcomes, encoding them somewhere internally, and behaving stochastically based on them. However, this is counterintuitively not necessarily the case. A fully deterministic strategy that encodes no rates whatsoever yields probability matching behavior. Consider a hypothetical rat running a series of T-maze trials, storing just 2 bits of information about only the previous trial: the first bit stores whether the rat turned left or right; the second whether it found food. If the rat was rewarded last trial, it deterministically goes the same direction; otherwise, it goes the opposite. The 2-bit memory is then overwritten with the new outcome. This hypothetical rat is not storing any rates and acts deterministically, yet its *win-stay, lose-shift* strategy yields probability matching (Robbins, 1952; Restle, 1962). This can be proven analytically or demonstrated with a simulation: Fig. 2 shows 30 simulated rats following this deterministic, non-rate-tracking strategy. This is not a claim that this is the strategy animals use when probability matching. Rather, it shows that *probability matching* is a description of a behavior not an implication of a probabilistic mechanism.

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These studies are sometimes discussed in relation to the observation that children often successfully learn and carry on variable linguistic patterns (Labov, 1994, 1989; Roberts, 1997). HZSL appeal to them in support of the “Law of Frequency Matching.” However, many of the experiments where humans probability match show use of a non-probabilistic win-stay, lose-shift strategy or an attempt to find structure in a sequence that the participant does not believe to be random (Gaissmaier & Schooler, 2008; Montag, 2021; Saldana *et al.*, 2022). Thus, Labov (1994, p. 583)’s recognition in saying “it is not a hypothesis that children do probability matching. It is simply a description of the observed facts” captures an important truth—that probability matching behavior does not imply probabilistic mechanisms—which H&L’s (p.91) conclusion that “it is evident from our experimental results that Hungarian-learning children ultimately develop a STATISTICAL MODEL of the lexicon” fails to.

Similarly, frequency effects are common in lexical processing. For instance, more frequent words are accessed more rapidly than less frequent words (Forster & Chambers, 1973). This is not in itself evidence that frequencies are being tracked. The effects are approximated by frequency-blind processes over lists of words, like the “move-to-front” algorithm, where an item is brought to the front of a list whenever it is accessed, or the “move-up” algorithm, which instead pushes each accessed item upward one position (Rivest, 1976; Sleator & Tarjan, 1985a,b; Murray & Forster, 2004). In both, frequency effects free-ride off the process without being encoded anywhere. Frequency effects also abound in the structure of morphological paradigms (e.g., Bybee 1985; Diessel 2007). Kodner (2023) demonstrated that these effects can emerge from generalization over *sparse paradigms* (Chan, 2008): empirically, children encounter only a fraction of the forms in most paradigms. Paradigm sparsity is related to frequency (children are more likely to know high-frequency forms), but indirectly via the rate of attestation in the input, not necessarily from the child tracking frequency.

These examples illustrate that frequency effects in language, including probability matching, do not imply probabilities being encoded in the lexicon, grammar, or otherwise, undermining H&L and HZSL’s conclusion of probabilistic grammars from the appearance of frequency matching.

## **2.2. The Problem of Individual vs. Group Variation**

Individual and group variation (or individual/group measures more generally) do not in general approximate each other. Gallistel *et al.* (2004) provide examples. They observe that in conditioning studies, it is common to plot a learning curve averaged over a group. In a classical conditioning study, pigeons might each be repeatedly presented

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an illuminated key as a conditioned stimulus, which is not intrinsically related to food but, when illuminated, is followed by the presentation of food. As the pigeon learns to associate the illumination of the key with the subsequent presentation of food, it will begin to peck at the key when it illuminates, as if it were the food. As the pigeons go through repeated reinforcement trials, the number of pecks per trial gradually increases, but only when averaging over pigeons. The individual pecks/trial for each pigeon are instead sudden, step-like curves. The “gradually increasing learning curve is an artifact of group averaging” (p. 13124) over the individual step-like curves, which vary in when the step occurs and in its magnitude.

H&L and HZSL consider that their results could reflect across-speaker variation rather than within-speaker variation, but dismiss the possibility despite their experiment including limited participant-level repeated measures.

H&L looked at individual-level responses for the categories of interest where each participant responded to multiple wugs. But they only included multiple wugs for four categories. For those, they had just 2-3 wugs per participant (see § 2.4). In those few cases, through a series of chi-squared tests, they did not find a statistical relationship between the multiple responses that participants gave to wugs within a category. Failing to reject the null does not prove the null (Gallistel, 2009). Regardless, variable responses would not imply *frequency matching*, which is the strictly stronger claim that each individual is not just variable but also *matching a specific rate of variants*. Moreover, a participant responding differently to two wugs from the same category of interest is not evidence of variation if the learner has internalized a different categorization of words than the experimenter has assumed, such that the two wugs are in fact importantly different and *not* in the same category from the participant’s perspective (see § 2.3). For instance, H&L use [ha:de:l] and [kole:n] as examples of BN<sub>mid</sub> wugs that were sometimes responded differently to. But we cannot know if a participant responding differently to the two is responding variably to BN<sub>mid</sub> wugs (what H&L assume) or if [ha:de:l] and [kole:n] are in fact importantly different to that speaker.

HZSL do not consider across-speaker variation for their own results. Discussing Ernestus & Baayen (2003), they assert that those results are more likely due to within-speaker variation than a “dialect split”—a false dichotomy. This is based on noting that some speakers in Baroni (1999)’s study of Italian [s] ~ [z] allophony and Ringen & Heinämäki (1999)’s study of Finnish vowel harmony sometimes responded to the same word differently across trials. As above, responding differently to the same word implies neither *frequency matching* specifically nor even variability (it could indicate guessing due to lack of a productive generalization, among other possibilities). Nor

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do studies on different patterns in different languages inform the likelihood of within-speaker variation in Dutch (or Hungarian for that matter).

In short, H&L and HZSL's confidence that their results reflect within-speaker variation is incompatible with their experiment design, which yields group-level values.

### 2.3. *The Problem of Potentially Wrong Categories*

Labov (1994, p. 585) notes that the first step, for child and linguist alike, in learning a variable pattern, “is to define the envelope of variation.” That is, delineating the set(s) of items subject to the variation is a prerequisite of matching the relative rates of variants. Co-opting this to lexical variation, if learners are matching type frequencies, they must identify a fraction, whose numerator and denominator are cardinalities of some sets, in order to match that frequency. For instance, the claim that learners match the rate of back-suffix-taking  $BN_{\text{high}}$  words assumes learners represent “ $BN_{\text{high}}$  words” as a type whose cardinality forms the denominator of a fraction. The learner must consider differences between  $BN_{\text{high}}$  words *unimportant*.

Because both the child and the linguist must define an envelope of variation, there is a risk that the linguist's envelope might deviate from the child's. If deviation occurs, the frequencies that the linguist looks for alignment on between a corpus and wug test will be different from those actually being matched, even if learners are indeed frequency matching. More concerning, such misalignment could lead categorical behavior to look like frequency matching.

Consider a simple non-linguistic example. A shape analyst pulls a collection of shapes from a bag, and finds they vary in color (*blue/gold*), as in Fig. (3A). The analyst believes circles and squares form important categories and that learners frequency match color within each. To test this, the analyst selects 100 test circles and 100 test squares (shape wugs) from the bag and asks a participant to predict the color of each. There are 16 circles, 10 of which are *blue*, so the analyst expects 62.5% of the circles to be predicted *blue*. Likewise, with 13/24 squares being *gold*, they expect 54.2% of the wug squares to be predicted *gold*.

It is possible that the participant thinks both the shape type and the dashed vs. solid perimeter are important attributes, instead forming the categorization in Fig. (3B). Moreover, suppose the participant *regularizes* (Hudson Kam & Newport, 2005) rather than *frequency matches* and thus predicts for all dashed circles *blue* and for all dashed squares *gold*. Since the wug and training shapes come from the same bag, on average 10/16 of the 100 wug circles (= 63) will be dashed, and the participant will respond

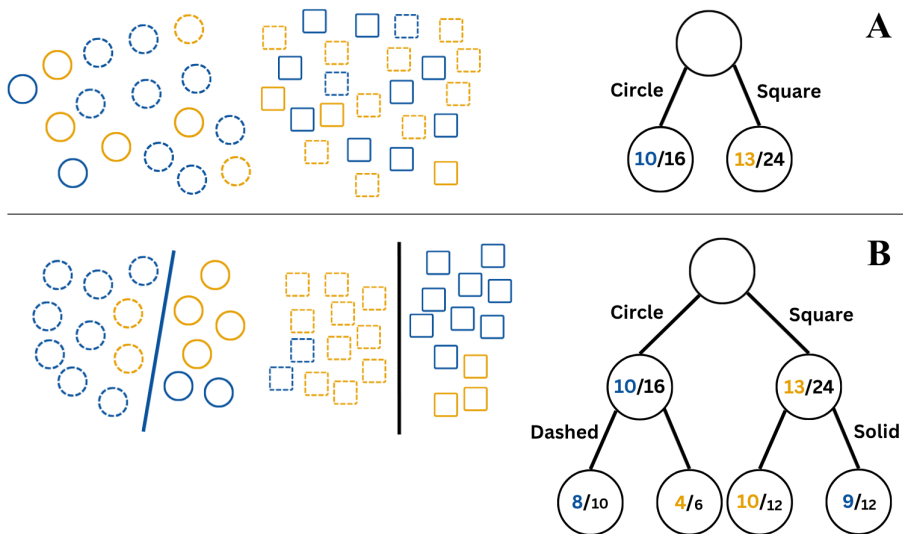


Figure 3: An example of how averaging over categories that may not reflect those internalized by learners can lead categorical behavior to look like frequency matching.

*blue* to all 63. The remaining 37 circles will be solid and the participant will respond categorically *gold*. Thus, 63% of the circle wug responses will be *blue*. Similarly, 12/24 of the 100 wug squares will be dashed, and the participant will respond to all 50 *gold* and all 50 solid squares *blue*. This yields 50% *gold* responses.

To the experimenter, who believes the participant is frequency matching circles and squares, 63% *blue* circles and 50% *gold* squares is effectively indistinguishable from the expected 62.5% and 54.2%, yet these numbers were produced by a categorical process. The appearance of frequency matching arises from smoothing over the categories that the learner actually used. It is thus not possible to evaluate evidence for frequency matching independently from a theory of the underlying categories.

While a simplified example, the results in § 4 suggest that there is reason to think that the categories H&L and HZSL average over are too coarse, analogous to this example.

#### 2.4. Could H&L and HZSL have Measured Individual Frequency Matching?

Momentarily putting aside these concerns, we can ask whether H&L and HZSL could have measured individual frequency matching under the assumption that the height  $\times$  count categories are the appropriate ones to consider.

H&L designed 30 wugs; each participant responded to one of two sets of 15. Example (4) shows the number of wugs (identical for both sets of 15) that each participant

responded to in the categories of interest (excluding the 4 total B, F, N, and NN wugs where lexical frequencies are effectively categorical).

| (4) | Category | Responses Per Participant | Measurable Fraction Back Values                   |
|-----|----------|---------------------------|---------------------------------------------------|
|     | Bi/i:    | 2                         | $\{0, \frac{1}{2}, 1\}$                           |
|     | Be:      | 2                         | $\{0, \frac{1}{2}, 1\}$                           |
|     | Bε       | 3                         | $\{0, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, 1\}$ |
|     | BNi/i:   | 1                         | $\{0, 1\}$                                        |
|     | BNe:     | 1                         | $\{0, 1\}$                                        |
|     | BNε      | 2                         | $\{0, \frac{1}{2}, 1\}$                           |

HZSL constructed 1602 wugs, but collected only 13 responses per participant. The specific wugs varied across participants, but the average number of wugs per category were identical to (4), except BNe averaged 1 wug rather than 2.

Thus, the measurable “fraction back” values is severely restricted. In a category like Be: with 2 responses per participant, each participant could either respond both back (Fraction Back = 1), both front (Fraction Back = 0) or one back and one front (Fraction Back = 0.5). For a category like BNe:, with only 1 response, the only measurable values would be categorically front or back. The largest category, Bε, with 3 responses, would still only have measurable values of  $\{0, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, 1\}$ .

Figure (4) shows, with black points, these measurable fractions for each category of interest. The orange line shows corresponding lexical frequencies. The black dashed line traces the measurable values that would be as close as possible to matching the lexical frequencies. This traces a flat line at 1.0 or 0.0. In other words, given the experiment design, the best match to lexical frequencies that an individual could achieve would be to respond categorically within every category. Thus, even if the height × count categories are appropriate, frequency matching and regularization, in which learners respond categorically with the dominant form in each category, are in principle indistinguishable in the experiment.

### 2.5. Summary of Critical Analysis

I first argued that probability matching behavior does not imply encoding of probabilities, but H&L and HZSL assume the appearance of frequency matching does imply a probabilistic grammar. Moreover, averaged group-level behavior is not indicative of individual behavior, but H&L and HZSL draw conclusions about individuals from averaged group-level behavior. Thirdly, evaluating the presence of frequency matching, even at the group level, requires an assumption about categories underlying the

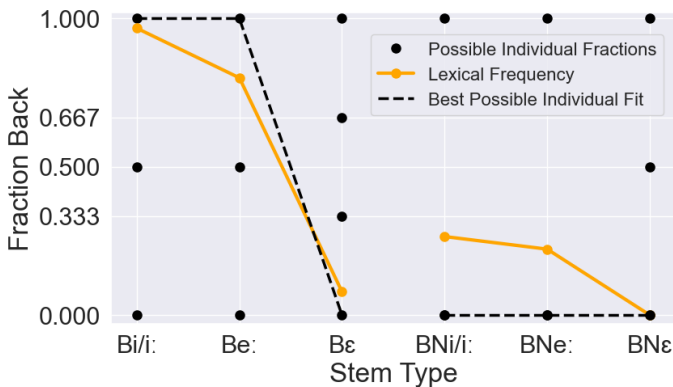


Figure 4: The best possible fit (dashed line) to corpus frequencies (orange) that an individual frequency matcher could achieve given the number of wugs per category of interest (dots) is exactly the response pattern achieved by responding categorically.

variation, which must be independently justified before drawing conclusions; H&L’s and HZSL’s analysis does not provide this justification. Putting these concerns aside, H&L and HZSL collected insufficient repeated measures from individuals to distinguish between stochastic and categorical behavior. Together, these suggest that the Frequency Matching Hypothesis, at least in the case of Hungarian vowel harmony, is on shakier ground than HZSL’s quip about the “Law of Frequency Matching” insinuates.

### 3. Proposal

Having argued that H&L’s and HZSL’s results did not provide, and could not have provided, evidence for the Frequency Matching Hypothesis (3), I turn to my proposal. Since variation requires identifying an envelope around it (Labov, 1994), the central question of lexical variation is a learning question: *how do learners discover the structural conditioning of lexical variation?*. Below, I propose an independently motivated approach to this question. Afterword (§ 3.4), I discuss how this learning theory supports a *variationist* (Weinreich *et al.*, 1968) theory of frequency matching, where H&L and HZSL results are indicative not of an idealized average speaker behaving stochastically but of variation across a population of learners, where the nature of that variation can be studied.

The theory is based on Belth (2026)’s learning-based account of phonological tiers. Belth argued that an independent psychological mechanism that learners bring to learning alternations is the ability to track dependencies between adjacent items. The evidence for this traces to statistical learning studies, where the ability to track adjacent

dependencies in a sequence develops earlier for children (by 8mo) (Saffran *et al.*, 1996, 1997; Aslin *et al.*, 1998) than the ability to track non-adjacent dependencies (around 15-18mo) (Santelmann & Jusczyk, 1998; Gómez, 2002; Gómez & Maye, 2005). Moreover, sensitivity to adjacency appears to be domain-general, since it is also observed with sequences of non-linguistics tones (Saffran *et al.*, 1999) and visual shapes (Fiser & Aslin, 2002) as well as in non-human primates (Hauser *et al.*, 2001).

Belth hypothesized that when learning phonological alternations, learners use this mechanism to attend to segments adjacent to the alternation and construct a generalization about the phonological conditioning of the alternation. If the resulting generalization is not sufficiently predictive of the alternation—formalized using the Tolerance/Sufficiency Principle (Yang, 2016)—learners construct a new representation. Importantly, learners use the same adjacency mechanism to construct the new representation: any currently representation-adjacent segment that is *not* a sufficient predictor of the alternation is excluded from the new representation (Heinz *et al.*, 2011). This process repeats recursively, forming a new adjacency-based generalization over the new representation. The resulting representations can be thought of as phonological tiers. Because the learner constructs both generalizations and representations via the same mechanism of tracking which alternation-adjacent segments are or are not sufficient predictors, it uses adjacency *all the way down*.

I apply this proposal, called D2L, to the present study as follows. The alternation is the choice between [-nɔk] and [-nɛk], with inputs like (5), which pair a stem with the suffix that the learner has heard the dative being formed with for that stem. A learner could encounter the same stem with both [-nɔk] and [-nɛk] (like the third form).

- |     |                           |               |
|-----|---------------------------|---------------|
| (5) | [mɔtʃkɔ] + [-nɔk]         | “cat”-DAT     |
|     | [kɒɲv] + [-nɛk]           | “book”-DAT    |
|     | [pɒrtner] + [-nɔk]/[-nɛk] | “partner”-DAT |

I will first show how D2L learns to predict [-nɔk] and [-nɛk] (§ 3.1) and then (§ 3.2) how this same mechanism can be extended to learn token-level variation, if present. While in principle important for genuine cases of token-level variation (e.g., Labov 1989; Roberts 1997), this extension ends up being unimportant to the particular case of Hungarian vowel harmony: in § 5.2 I show that stems occurring with both suffixes are remarkably rare in Hungarian and are probably more indicative of across-speaker variation than of genuine token-level variation.

### 3.1. Adjacency and Prediction in D2L

Following Belth, I hypothesize that learners attempt to form a generalization about the choice of suffix based on representation-adjacent segments, which on an initial linear representation are stem-final segments.<sup>1</sup>

Let us call the set of tier-adjacent segments,  $A$ . D2L selects the subset of segments of  $A$  that are *sufficiently predictive* of  $[-n\sigma k]$  or  $[-n\epsilon k]$ , as measured by the TSP (Yang, 2016) as in (6).

- (6) A tier-adjacent segment,  $s \in A$ , is **sufficiently predictive** if, of the  $n_s$  stems that end in that segment on the current tier, the most common following suffix ( $[-n\sigma k]$  or  $[-n\epsilon k]$ ) has type frequency  $m_s$ , and  $n_s - m_s \leq \frac{n_s}{\ln n_s}$ .

I will use  $\theta_n = \frac{n}{\ln n}$  as a shorthand for the threshold at  $n$ . Let the subset of tier-adjacent sufficient predictors be  $P_A = \{s \in A : \text{TSP}(n_s, m_s)\}$ , where  $\text{TSP}(n_s, m_s) = \text{TRUE}$  if  $n_s - m_s \leq \theta_{n_s}$  and  $\text{FALSE}$  otherwise. The non-sufficient predictors are then  $D_A = A - P_A$  (where ‘ $-$ ’ is the set difference).

As an example, consider a learner who has learned the  $[\sigma]$ -final words in (7a) and the  $[\epsilon]$ -final words in (7b). All  $m_\sigma = 3$  of the  $n_\sigma = 3$  of the former take  $[-n\sigma k]$ , so  $[\sigma]$  is a sufficient predictor of  $[-n\sigma k]$ . Similarly,  $m_{\epsilon} = 4$  of  $n_{\epsilon} = 7$  of the latter take  $[-n\epsilon k]$ ; since  $7 - 3 \leq \theta_7$ ,  $[\epsilon]$  is a sufficient predictor of  $[-n\epsilon k]$ . Both are added to the set  $P_A$ .

- (7) a.  $[\text{m}\sigma\text{t}]\widehat{[\text{k}\sigma]} + [-n\sigma k]$  “cat”-DAT  
 $[\text{f}\sigma\text{r}]\widehat{[\sigma]} + [-n\sigma k]$  “fate”-DAT  
 $[\text{old}\sigma\text{l}\sigma] + [-n\sigma k]$  “side”-DAT  
 b.  $[\text{byd}]\widehat{[\text{z}\epsilon]} + [-n\epsilon k]$  “budget”-DAT  
 $[\text{d}:\text{be}:\text{tse}:\text{e}] + [-n\epsilon k]$  “alphabet”-DAT  
 $[\text{ese}:\text{e}] + [-n\epsilon k]$  “essay”-DAT  
 $[\text{fyle}:\text{e}] + [-n\epsilon k]$  “ear”-DAT  
 $[\text{k}\sigma:\text{ve}:\text{e}] + [-n\sigma k]$  “coffee”-DAT  
 $[\text{p}\sigma\text{r}\sigma:\text{de}:\text{e}] + [-n\sigma k]$  “parade”-DAT  
 $[\text{turne}:\text{e}] + [-n\sigma k]$  “tour”-DAT

Suppose another learner knows the same words and also (8).

- (8)  $[\text{fin}\sigma:\text{le}:\text{e}] + [-n\sigma k]$  “finale”-DAT

Because this learner knows  $n_{\epsilon} = 8$   $[\epsilon]$ -final words and  $4 > \theta_8$ ,  $[\epsilon]$  is neither a sufficient predictor of  $[-n\epsilon k]$  nor of  $[-n\sigma k]$  for this learner, so it is added to  $D_A$ .

<sup>1</sup>D2L also considers right-adjacent segments and discovers which context is (not) predictive, but the right context is obviously not predictive in this case, so for clarity I present the model in terms of just the left-adjacent context.

After constructing the sets  $P_A$  and  $D_A$ , D2L evaluates the predictors  $P_A$  together under the TSP. Each predictive segment  $s \in P_A$  makes  $n_s$  predictions,  $e_s$  of which are incorrect. I will use the non-subscripted  $n$  and  $e$  to refer to the total predictions and incorrect predictions,  $n = \sum_{s \in P_A} n_s$  and  $e = \sum_{s \in P_A} e_s$ , respectively.

D2L first checks that  $P_A$  covers a necessary number of words: of the  $t$  total words,  $n$  of which receive a prediction from  $P_A$ , it checks  $t - n \leq \theta_t$ . Following prior work on D2L, the segments that do not receive a prediction from  $P_A$  form an *elsewhere condition*, a sufficient number of which are required to take one suffix (Belth, 2026; Yin, 2026). In theory, this allows for capturing default forms in vowel harmony, where words with only transparent vowels mostly take the same suffix. In practice though, an elsewhere condition was never formed in the experiments, as NN words were all subsumed under some sufficient predictor of  $[-n\epsilon k]$  (e.g., the left word/tier edge, which is reached if no stem-internal segment is a sufficient predictor, is itself typically a sufficient predictor of  $[-n\epsilon k]$ ).

If the predictors,  $P_A$ , cover a necessary number of the words, the predictors are required to be collectively sufficient:

- (9) Of the  $n$  predictions  $P_A$  makes,  $e$  of which are incorrect,  $P_A$  is a sufficient predictor if  $e \leq \frac{n}{\ln n}$ .

If in this way  $P_A$  constitutes a necessary and sufficient predictor of  $[-n\epsilon k]$  vs.  $[-n\epsilon k]$ , D2L stops, having constructed a sufficient grammar. Otherwise, it forms a new representation for each word by projecting the segments,  $D_A$ , that are not sufficient predictors. For instance, if  $D_A = \{j, j, l, m\}$ , then the words in (10) all become  $[e:]$ -tier-adjacent on the next iteration.

- (10)  $/s\epsilon me:j/ \rightarrow <s\epsilon e:>$   
 $/e\jmath \epsilon me:j/ \rightarrow <e\jmath \epsilon e:>$   
 $/\omega tse:l/ \rightarrow <\omega tse:>$

D2L then computes the set of adjacent sufficient predictors  $P_A$  on the new tier, and checks whether it is a necessary and sufficient predictor. This process repeats iteratively/recursively, accumulating tier-adjacent segments that are non-sufficient predictors into  $D_A$  at each iteration, until  $P_A$  becomes productive under the TSP. This iterative process is summarized in Fig. 5. The double circle represents that state where a deletion set  $D_A$  supporting a necessary and sufficient predictor  $P_A$  has been constructed, forming a learned grammar, which I will call  $G = (D_A, P_A)$ .

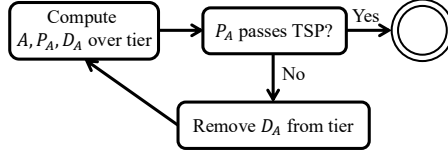


Figure 5: Diagram of D2L.

### 3.2. Token-Level Variation in D2L

I extend D2L to handle token-level variation using [Yue \(2026\)](#)’s approach. [Yue](#) proposes that variation is learned when the input distribution supports multiple productive generalizations. Here, if a word is attested in the learner’s input with both  $[-n\text{ɔ}k]$  and  $[-n\text{ɛ}k]$ , that form contributes to the count of both  $[-n\text{ɔ}k]$  words and  $[-n\text{ɛ}k]$  words for whichever segment is tier-adjacent at a given iteration. If the tier-adjacent segment is a sufficient predictor of only one suffix, future stems with the same tier-adjacent segment are predicted to take that suffix. But if it is a sufficient predictor of *both* suffixes, then future stems with that tier-adjacent segment are predicted to be variable, taking either suffix. Such tier-adjacent segments are predictors of variation.

If variation is learned, D2L probability matches over the  $[-n\text{ɔ}k]$  and  $[-n\text{ɛ}k]$  tokens of all types tier-adjacent to the predictor. For present purposes, probability matching was implemented by counting the number of tokens (summed over types) with  $[-n\text{ɔ}k]$  in the learner’s input, out of the total tokens for tier-adjacent types. This is not meant as a commitment to tracking of rates as the mechanism underlying probability matching (see § 2.1 for discussion of non-quantity-tracking mechanisms); such an algorithmic description ([Marr, 1982](#)) of probability matching is beyond the paper’s scope.

For example, suppose there are  $n_{e:} = 8$  ( $\theta_8 = 3.8$ )  $[e:]$ -tier-adjacent items, 7 of which occur with  $[-n\text{ɛ}k]$  and 5 of which occur with  $[-n\text{ɔ}k]$ —4 occurring with both. Because both  $8 - 7 \leq \theta_8$  and  $8 - 5 \leq \theta_8$ ,  $[e:]$  is a sufficient predictor of both  $[-n\text{ɛ}k]$  and  $[-n\text{ɔ}k]$ . Consequently,  $[e:]$  predicts variation. The probability of  $[-n\text{ɔ}k]$  then matches the number of  $[-n\text{ɔ}k]$  tokens of the 5  $[-n\text{ɔ}k]$ -attested types, divided by the total number of tokens for all 8 types. In the experiments below, token-level variation only arose for one segment on 1/100 simulations on one dataset (see § 4.2 and § 5.2).

### 3.3. Structural Conditioning in D2L

The learned grammar,  $G$ , partitions the input into a set of equivalence classes, each determined by a tier-adjacent sufficient predictor. The notation  $w_i \sim_G w_j$  means  $w_i$  and  $w_j$  are in the same equivalence class (11).

- 
- (11)  $w_i \sim_G w_j$  if they have the same tier-final segment after projection under the learned grammar  $G$ .

$G$  makes the same prediction of suffix for any two words in the same equivalence class. Thus, the set of tier-adjacent segments constitutes D2L’s learned structural conditioning of lexical variation.

### 3.4. A Variationist Theory of Frequency Matching

D2L is a model of an individual learner. Each learner constructs a grammar, where the equivalence classes produced by the tier-final segments are that learner’s theory of the structural conditioning of lexical variation. Whether a learner preserves a segment,  $s$ , on the tier as a sufficient predictor depends on the values  $n_s$  and  $m_s$ , which are type frequencies computed over that learner’s lexicon. Consequently, learners with different vocabularies—even subtly different—might learn slightly different structural conditioning. Below I demonstrate that the appearance of frequency matching in H&L and HZSL can emerge from variation across a group of such learners (§ 4). I call this group-level proposal  $\text{D2L}^\pi$ , where  $\pi$  stands for a *population* of learners:

$\text{D2L}^\pi = \{\text{D2L}_1, \text{D2L}_2, \dots, \text{D2L}_k\}$ .

Because the structural conditioning is interpretable as the equivalence classes formed by the tier-adjacent predictors (§ 3.3), it is possible to investigate the amount and character of the variation, which I will do in § 4.4. As expressed in the evolutionary biologist Ernst Mayr (1982, p. 47)’s celebrated history and philosophy of biology, “The most interesting parameter in the statistics of natural populations is the actual variation, its amount, and its nature” rather than the average, which is an idealization.

## 4. Frequency Matching Results

The first experiments aim to show that  $\text{D2L}^\pi$ , where variation emerges from differences in the structural conditioning learned by speakers, leads to the appearance of frequency matching in H&L/HZSL’s wug tests. I introduce the data and setup (§ 4.1) before demonstrating that even though each D2L model in  $\text{D2L}^\pi$  behaves categorically, the appearance of frequency matching occurs when averaging over models and the H&L height  $\times$  count categories (§ 4.2). I then quantitatively evaluate the fit of  $\text{D2L}^\pi$ ’s predictions to participant wug responses (§ 4.3). I then probe the richness and limits of variation predicted by  $\text{D2L}^\pi$  (§ 4.4).

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## 4.1. Experiment Setup

I discuss how I modeled a group of Hungarian learners (§ 4.1.1), what data I used for evaluation (§ 4.1.2), and how I compared to H&L/HZSL’s probabilistic grammar approach (§ 4.1.3).

### 4.1.1. Simulating Hungarian Learners

Modeling a group of Hungarian speaker/learners with D2L <sup>$\pi$</sup>  requires simulating variable vocabularies. It is common practice to sample words from a corpus, weighted by token frequency since learners are more likely to know higher frequency words (Goodman *et al.*, 2008; Swingley & Humphrey, 2018; Belth *et al.*, 2021; Kodner *et al.*, 2025), and repeat for a number of pseudorandom seeds to simulate multiple learners. This produces variation in the particular words that a learner knows. I followed this basic sampling procedure, but introduced some variation on the *number* of words they know. The simulation procedure is summarized in (12).

(12) **Input:** Dataset

- Sample number,  $N$ , of datives known:  $N \sim \text{Normal}(\mu, \sigma)$
- Sample  $N$  words weighted by token frequency
- Train a D2L model on  $N$  words
- Wug test trained model

Repeat 100 times

This simple simulation is only a process to generate variation in lexicons, and should not be taken as a theory of lexical development. Variation in lexicon size is simulated with a Gaussian (normal) distribution: for each simulated learner, a vocabulary size is sampled  $N \sim \text{Normal}(\mu, \sigma)$ . For each simulated learner/speaker, the procedure samples  $N$  words from the dataset weighted by their token frequency. It then trains a D2L <sup>$\pi$</sup>  model on each sample and wug tests each resulting model.

Lacking estimates for the number of datives Hungarian learners might know by the time they converge to a stable grammar, I chose  $\mu = 4000$  and  $\sigma = 500$ , for two reasons. First, it is roughly half the size of HZSL’s dataset. A relatively small number of stems are BN/BNN (5% of HZSL’s dataset), so a value too small might provide too little information about the lexical variation. But a value too large relative to the dataset size would saturate the dataset, artificially limiting the amount of vocabulary variation via the artificial limit to the dataset size. Second, after running the simulation (12) for  $\mu \in \{2000, \dots, 7000\}$  for 100 seeds each, I evaluated the accuracy of each model’s predictions on the dataset forms not seen during training. This test accuracy largely converged by 3000-4000, suggesting that input beyond this is of diminishing

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value. Regardless, the fit to wugs (§ 4.3) is comparable across values, so long as they are sufficiently large but small enough to not artificially saturate the dataset.

#### 4.1.2. Datasets

Many of the results below are based on H&L’s web-constructed dataset of Hungarian datives, as refined by HZSL. They created this dataset by taking a word list and searching Google for the number of hits for each word suffixed with *-nak* and *-nek*. The dataset contains 8915 stems. I will refer to this dataset as HAYES.

As a second dataset, I used the morphologically- and token-frequency-annotated word list of the Hungarian National Corpus (Váradi, 2002), which was constructed from 187.7 million words of Hungarian text, from a variety of genres, and is available online.<sup>2</sup> I extracted all stems occurring in dative form that did not contain special characters, which resulted in a list of 53800 stems occurring in dative form (*-nak*, *-nek*, or both). I will refer to this dataset as HNC.

While I most centrally use HAYES for continuity with prior work, the HNC is important for three reasons. First is robustness. In § 4, I demonstrate that  $D2L^\pi$ ’s fit to human wug test results generalizes across datasets, an encouraging result for the proposal. Second, in § 5.2 I demonstrate that the way the HAYES dataset was created (see below) prevents accurate reasoning about token-level variation (*vacillators*), whereas the HNC enables doing so. Lastly, while both corpuses have similar statistics for BN/BNN forms, some interesting discrepancies align conspicuously with the variation predicted by  $D2L^\pi$  (§ 4.4), thereby coincidentally serving as further evidence for my proposal.

Because the HAYES dataset was constructed from a word list, the number of types in the corpus is capped by the list size. The token frequencies were estimated from the internet, where the number of types is much larger than H&L’s list. This distorted the type/token ratio. In effect, the tail of the Zipfian distribution (low-frequency types) of the underlying internet corpus was truncated, producing highly inflated token frequency estimates in the tail of the resulting HAYES dataset. I generated rank-frequency plots for the HAYES corpus (6a) and the HNC (6b) in log-log scale in Figure (6). Even though the HAYES dataset has an order of magnitude fewer types than HNC (8.9K vs. 53.8K), it has an order of magnitude more tokens (8.4M vs. 968K). This massive inflation of the token frequency of the tail leads the dataset to not be Zipfian. In a Zipfian distribution, the tail is characteristically approximated by a line in log-log scale like HNC in Fig. 6b and the famous Brown corpus (Francis & Kučera, 1979) of English in Fig. 6c. The tail of the HAYES distribution is yanked upward, producing a hump. This

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<sup>2</sup>[Link to frequency list download.](#)

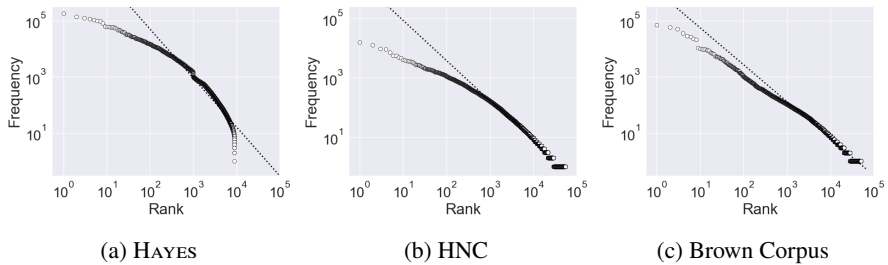


Figure 6: Rank-Frequency Plots.

greatly inflates the number of chances to observe a given type with both suffixes in HAYES. Because both type and token frequencies come from the same source in HNC, it does not suffer in the same way. This has implications for § 5.2.

#### 4.1.3. Probabilistic Grammar Comparison

To compare to the probabilistic grammar proposal of H&L and HZSL, I used the MaxEnt (Goldwater & Johnson, 2003) grammar constructed by HZSL. H&L used the Gradual Learning Algorithm (Boersma *et al.*, 1997; Boersma & Hayes, 2001), which learns a Stochastic OT grammar rather than a MaxEnt Harmonic Grammar, but HZSL’s MaxEnt grammar added additional constraints, making it a more developed instantiation of their proposal. I used the weights reported by HZSL (p. 848), which they produced by fitting the model to the HAYES corpus, to generate a probabilistic prediction for each wug. For comparison to  $D2L^\pi$  when trained on HNC, I generated a secondary set of constraint weights by fitting the same MAXENT model to the HNC.

#### 4.2. The appearance of Frequency Matching

I simulated, via the procedure above on the HAYES dataset, 100 Hungarian learners with  $D2L^\pi$  and wug tested each. The tier-adjacent sufficient predictors were always categorical (no token-level variation), with one exception: on just 1 of 100 seeds, [f] was a sufficient predictor of both [-nɔk] and [-nɛk] (6/10 [f]-tier-adjacent types occurred with [-nɔk] and 6/10 with [-nɛk]; 2 with both). This was a marginal case, since it covered only 10 types, occurred on only 1 seed, and did not apply to any wugs. Moreover, when trained on HNC in subsequent sections, all predictors were categorical (I show in § 5.2 that the HAYES data inflates the amount of token-level variation in the input). Thus, each  $D2L$  makes fully categorical predictions. I computed, for each of H&L’s categories (count  $\times$  height and F, N, and B), the fraction of model wug responses in each that were back [-nɔk]. For each category, I also computed the probability of

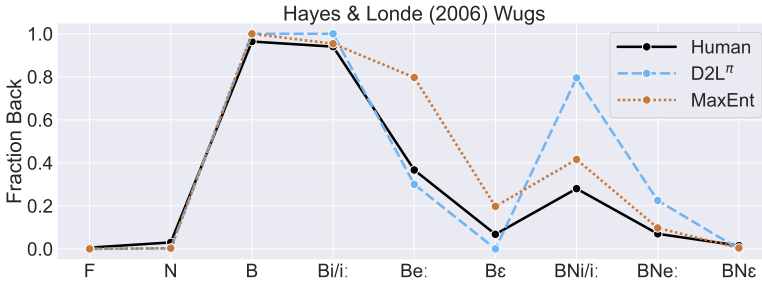


Figure 7: The appearance of frequency matching with H&L’s wug test results.

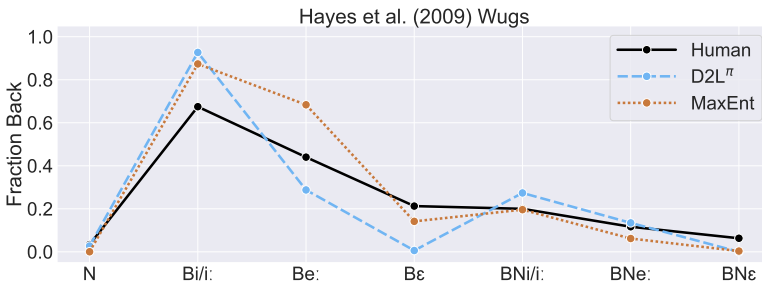


Figure 8: The appearance of frequency matching with HZSL’s wug test results.

back response that MAXENT predicts. I compared these predictions to the fraction of human back responses in each category. The results are plotted in Figs. 7-8: black is the humans’ values, blue D2L<sup>π</sup>’s, and orange MAXENT’s.<sup>3</sup>

Both models visually approximate the human rates of responses. The human results are aggregates across the participants and D2L<sup>π</sup>’s results are aggregates across D2L models. D2L<sup>π</sup> shows the appearance of frequency matching. Thus, the appearance of frequency matching in H&L and HZSL’s wug tests can emerge from averaging over individual learners, each behaving categorically, and over categories that may not reflect the structural conditioning internalized by speakers.

#### 4.3. Fit to Wugs

To evaluate model quantitative fits to wugs, I used several metrics:

- **MSE:** the mean squared error between the fraction of back responses in the wug test and the fraction predicted by each model, ranging in [0, 1]. I report 1 – MSE

<sup>3</sup>Fig. 8 has no B and F values because HZSL had no B or F wugs.

Table 1: Quantitative comparison between model predictions and wug test results. The top rows show models trained on the HAYES dataset; the bottom when trained on the HNC. Across metrics and datasets, D2L $^\pi$  matches the participant responses as well as MaxEnt does. No differences are statistically significant.

|       |            | Hayes & Londe (2006) |       |       |       | Hayes <i>et al.</i> (2009) |       |       |       |
|-------|------------|----------------------|-------|-------|-------|----------------------------|-------|-------|-------|
|       | Model      | 1-MSE                | $r$   | Acc   | Sim   | 1-MSE                      | $r$   | Acc   | Sim   |
| HAYES | D2L $^\pi$ | 0.976                | 0.928 | 0.933 | 0.852 | 0.813                      | 0.446 | 0.778 | 0.779 |
|       | MaxEnt     | 0.967                | 0.917 | 0.867 | 0.808 | 0.839                      | 0.527 | 0.785 | 0.771 |
| HNC   | D2L $^\pi$ | 0.965                | 0.902 | 0.933 | 0.834 | 0.816                      | 0.465 | 0.785 | 0.777 |
|       | MAXENT     | 0.976                | 0.901 | 0.867 | 0.797 | 0.836                      | 0.462 | 0.778 | 0.756 |

so that larger values correspond to better agreement between model predictions and human responses.

- **Pearson’s  $r$ :** the correlation between the fraction of back responses observed for each wug and that predicted by each model. I report this for continuity with H&L/HZSL, who used it as the sole measure of model agreement, but note that most wugs have nearly categorical responses, so the majority of values being correlated to are binary (responses on HZSL’s wugs are 99.9% binary). Since MAX-ENT and D2L $^\pi$  output probabilities (D2L $^\pi$  via the fraction of models predicting back), this changes the interpretation of Pearson’s  $r$ : correlating probabilities to binary values measures discrimination, not calibration of probabilities.
- **Acc:** the accuracy between the wug responses and model predictions after binarizing the fraction back values. Where responses are binary, this measures the fraction of wugs for which the response the model predicted to be most probable was that actually observed.
- **Sim:** the similarity between the participant response profiles predicted by models and those observed. Each participant has a response profile: a binary vector of responses where 1 vs. 0 captures a back vs. front choice. H&L’s participants each responded to 15/30 wugs and HZSL’s each responded to 13/1602. Likewise, each model produces a binary vector of predictions (since MAXENT embodies the idealized average speaker, I generated 100 vectors by sampling each wug choice weighted by MAXENT probability). For every model-participant pair, I computed the Hamming distance between the participant vector and the subvector of the model corresponding to the wugs that participant responded to, which measures the fraction of bits that differ between the vectors out of the total number of bits

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compared (ranging in  $[0, 1]$ ). I convert distance to similarity by subtracting from 1. This yields a distribution of pairwise model-participant similarities, where a distribution closer to 1 indicates that predicted model response profiles look more like those observed. The mean of the distribution is Sim.

Table 1 shows the results. The first row shows  $D2L^\pi$ , consisting of 100 D2L models trained on the HAYES dataset. The second row shows MAXENT. The third and fourth rows show the same, but trained on the HNC, for evaluation of robustness across datasets. Metrics are reported for the H&L and HZSL wugs separately.

Across metrics and wug tests and regardless of training dataset,  $D2L^\pi$  achieves at least as good match as MAXENT to both the H&L and HZSL wugs. No difference between models on  $1 - \text{MSE}$ ,  $r$ , or Acc was statistically significant under a bootstrap significance test. Due to sparsity (only 13-15 responses per participant) it was not possible to bootstrap test Sim.

Since each D2L model in  $D2L^\pi$  makes categorical predictions,  $D2L^\pi$ 's performance shows that H&L and HZSL results can be accounted for by variation across speakers, undermining H&L and HZSL's conclusion that their results are indicative of individuals frequency matching.

#### 4.4. *The Nature of Variation in $D2L^\pi$*

H&L and HZSL's view of variation is that individuals match type-level frequencies, as expressed by a probabilistic grammar, which is presumed to be internalized mostly identically by each speaker. H&L (p. 74) are clear that they believe the variation is within not across individuals: "We conclude that idiolect variation played at most a minor role in our results. The level of variation is not between individuals, but within the individual: when confronted with a novel wug stem, each speaker behaved stochastically, in a way that matched the frequencies of the lexicon." Much of § 2 was dedicated to demonstrating the untenableness of this conclusion.

In contrast, my proposal places the variation across individuals. It is *variationist* (Weinreich *et al.*, 1968) in the sense that it recognizes variation as "the irreducible essence" (Gould, 1985, p. 140). Because the H&L/HZSL view of variation rejects substantial inter-speaker variation and treats the grammar as an average over individuals, it is a *non-variationist* view, in which the abstraction (the average) not the variation is the essence. In modeling participant averages, H&L and HZSL follow Quetelet, who hoped to characterize the "average man" (Mayr, 1982, p.47).

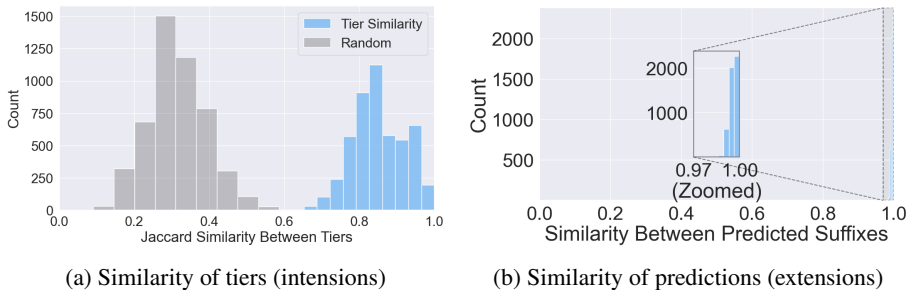


Figure 9: Grammars learned by different  $D2L^\pi$  models differ meaningfully intensionally (a), but are remarkably consistent extensionally for the words Hungarian speakers use most often (b).

Because of  $D2L^\pi$ 's variationist nature, it is possible to investigate the richness and limits of the variation. There are broadly two ways of thinking about variation across  $D2L^\pi$  models: the *intension* or form of the grammar and its *extension* or predictions. Since the set of tier-final segments partitions the input space into equivalence classes, an important aspect of the intension of a  $D2L$ -learned grammar is the set of segments preserved on the tier, that is its learned *tierset*. I computed the Jaccard similarity between the tiersets of each pair of models ( $D2L_i, D2L_j$ ) in  $D2L^\pi$ . Jaccard similarity measures the cardinality of the intersection divided by the cardinality of the union, and ranges from 0 (non-overlapping tiersets) to 1 (identical tiersets). The distribution of pairwise similarities between the 100 models is shown in blue in Fig. 9a. For comparison, the gray distribution shows the pairwise Jaccard similarities after generating for each model a random tierset of the same size as that actually learned. The agreement across models is substantial, but also falls meaningfully short of total agreement.

However, if we turn to the *extensional* difference, the nature of the variation comes into focus. I computed the pairwise Hamming similarities between the predictions of each pair of models on the 1K most frequent datives. These are the datives Hungarian speakers are most likely to use day-to-day. A value of 1 means both models predict identical suffixes for all 1K datives; a value of 0 indicates total divergence. This is plotted in Fig. 9b. All values are  $\approx 1$ , such that the extensional variation is barely visible with the full  $[0, 1]$  range for the  $x$ -axis. The inset zooms in on the  $[0.97, 1.0]$  range; most values still hug 1.0. Under the  $D2L^\pi$  proposal, even though learners converge to intensionally different grammars (Fig. 9a), in day-to-day life, they are unlikely to encounter many situations where their grammars clash (Fig. 9b).

Of great interest are the neutral vowels. Figure (10) shows for each neutral vowel the fraction of models where the vowel was preserved on the tier and remained opaque (i.e.,

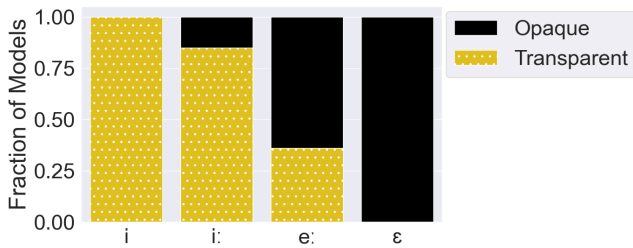


Figure 10: The fraction of  $D2L^\pi$  models in which each neutral vowel is retained on the tier as a sufficient predictor (opaque) or is not (transparent). The vowels [i:] and [e:] show the greatest variation.

predictive of [-nɛk]) in harmony (black) compared to the fraction of models where it was not, making it transparent to harmony (gold). The height effect emerges: moving left to right is a move from higher to lower vowels, and the fraction of models where each vowel is transparent decreases for the lower vowels. Interestingly, an additional difference arises: the short [i] and long [i:] show an important difference, where [i:] but not [i] is sometimes preserved as opaque. I am not aware of prior reports of this difference. The same ordering is observed for  $D2L^\pi$  trained on the HNC.

The fact that  $D2L^\pi$  predicts the greatest variation for [i:] and [e:] aligns with a difference between the HAYES and HNC datasets. Figure 11 shows the absolute difference between the fraction back values of H&L’s categories of interest, but divides  $BN_{high}$  into  $BN_i$  and  $BN_{i:}$ . While most categories have nearly identical fraction back values in both datasets, the  $BN(N)$  stems that end in [i:] and [e:] show the greatest variation across datasets. This provides support for  $D2L^\pi$ ’s picture of variation and is an example of H&L’s categories of interest being too coarse, raising the concern that averaging over them might smooth over important differences between wugs (§ 2.3).

In  $D2L^\pi$ , the envelope of lexical variation, when considered variationally across learners, is a porous layer around a robust core. The long vowels [i:] and [e:] fall near this porous layer, while [i] and [ε] are in the robust core. Due to falling near the porous layer, [i:] and [e:] are susceptible to diachronic instability, where they may become more opaque over time due to some learners failing to remove them from the tier, instead leaving them as sufficient predictors of [-nɛk]. I discuss this possibility in § 5.3.

The count effect also emerges from  $D2L^\pi$ , as seen in Fig. 12, which shows the fraction of back predictions for BN vs. BNN stems, averaging across  $D2L^\pi$  models. The reason is that any segment between the back vowel and the stem’s right edge that is preserved on the tier as a sufficient predictor of [-nɛk] will predict [-nɛk]. Because the B vowel is further inwards in BNN stems than BN stems, there are simply more such

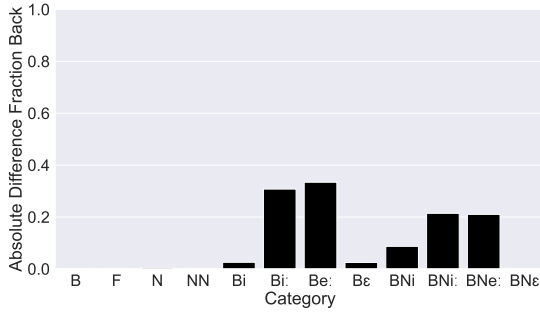


Figure 11: The absolute differences between the Fraction Back values in the HAYES corpus and those in the HNC. Most are highly similar, but the BN(N) stems ending in [i:] and [e:], which D2L predicts to show greatest variation, show the largest difference.

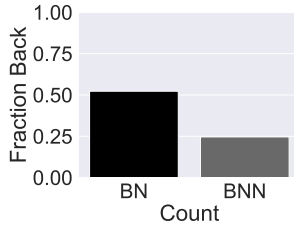


Figure 12: The count effect emerges from D2L $^{\pi}$ .

chances for BNN stems. Thus, both the height and count effects emerge from D2L $^{\pi}$ , as well as more subtle effects like a “length effect” between [i] and [i:].

## 5. Arguments for the Proposal

I turn now to positive arguments for D2L $^{\pi}$ ’s variationist proposal.

### 5.1. The Argument from Psychological Mechanism and Cross-Linguistic Variation

The ability to identify dependencies between adjacent items is one of the most celebrated findings of psycholinguistics (Saffran *et al.*, 1996, 1997; Aslin *et al.*, 1998; Hauser *et al.*, 2001; Fiser & Aslin, 2002). D2L is based on this mechanism, using it both to construct generalizations and, when they fail, to construct new representations. It assumes nothing specific to Hungarian. Belth (2023a, 2026) demonstrated that D2L learns Turkish and Finnish vowel harmony, producing the same kinds of errors that children produce during acquisition (Altan, 2009) as well as mirroring human generalization patterns in experimental studies (Finley, 2011, 2015; McMullin & Hansson,

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2019; Belth, 2023b, 2025). The model is not restricted to harmony, as local alternations (e.g., local assimilation) are learned by the same adjacency mechanism, with no tier being constructed.

In contrast, the HZSL MaxEnt model has 13 constraints, several hand-tailored for Hungarian vowel harmony (13).

- (13) Agree(back, local)
- Agree(back, nonlocal)
- Agree(front rounded, local)
- Agree(front rounded, nonlocal)
- Agree(front, local)
- Agree(nonhigh front, local)
- Agree(low front, local)
- Agree(double front, local)
- Use Front / bilabial \_\_
- Use Front / [+cor,+son] \_\_
- Use Front / sibilant \_\_
- Use Front / CC \_\_
- Use Front / [C<sub>0</sub>i:C<sub>0</sub>] \_\_

For instance, the constraint set separates Agree(front) across 6 constraints over different levels of height and different numbers of vowels to capture the height and count effects. Because constraint weights are the parameters of a MaxEnt model (a form of logistic regression where the constraints are predictors), this builds these effects into the model parameters. It also means that to learn vowel harmony in another language, a different set of hand-crafted constraints would be needed.

D2L is a psychologically-motivated alternation learner that here learns the structural conditioning of lexical variation. The height and count effects (§ 4.4), as well as the appearance of frequency matching when averaging over learners (§ 4.2), emerge without any hand-tailoring to Hungarian. MAXENT requires hand-crafted constraints that build in the structural conditioning and a logistic function to convert predictions to probabilities. It can thus not non-circularly explain the height and count effects or the appearance of frequency matching.

## 5.2. The Argument from (Lack of) Token-Level Variation

H&L and HZSL discuss *vacillators*, which were words attested with both suffixes in their corpus. They took these vacillators to indicate true token-level variation (p.83),

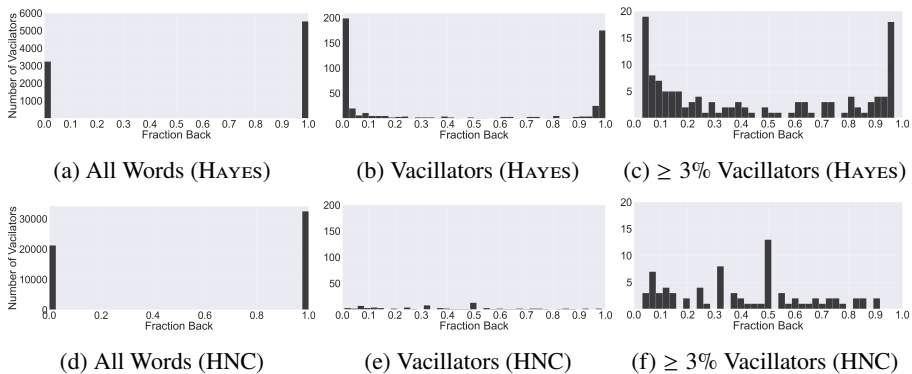


Figure 13: The distribution of vacillators in HAYES and HNC. Token-level variation is highly restricted.

internalized by speakers. They suggest that this could be represented in the lexicon via lexical learning. Other work has proposed lexically-indexed constraints to capture this sort of lexically idiosyncratic behavior (Coetzee, 2009; Pater, 2009; Zuraw, 2010; Linzen *et al.*, 2013; Moore-Cantwell & Pater, 2016; Zymet, 2018; Hugtho *et al.*, 2019).

Under the present proposal, vacillators in Hungarian reflect inter-speaker variation, rather than genuine token-level variation. To compare these views, Fig 13a provides a histogram of the number of datives in HAYES with different fraction back values. The behavior overall seems clearly categorical. Fig. 13b is the same plot, but only for the subset of items that occur at least once with both  $[-n\partial k]$  and  $[-n\epsilon k]$ —that is, the vacillators. Note the dramatic drop in the y-axis scale, which indicates this is a small subset of the words (5.75%). Moreover, the distribution is still dominated by two bars at  $\approx 0.0$  and  $\approx 1.0$ , indicating that even the so-called vacillators are nearly categorical. In some of their calculations, H&L filtered down to only the vacillators that occur at least 3% of the time with both suffixes (1.44% of datives); Fig. 13c gives the resulting histogram. Even then the token-level variation is extremely limited.

Moreover, the type-token ratio distortion caused by the creation process of the HAYES corpus (see § 4.1.2) greatly inflates the fraction of datives that appear to be vacillators. Figures 13d-13f are the analogous plots for the HNC. The y-axes for Figs. 13e-13f match those for Figs. 13b-13c for comparison. Even though HNC is over 6× the size of HAYES, there are fewer vacillators in HNC.

In HAYES, where the type-token ratio is so distorted, 512 of the 8,915 (5.74%) datives occur at least once with both suffixes (though only 1.44% occur at least 3% of the time with both). In the HNC, where the type-token ratio is reliable, only 82 of the 53,800 datives (0.15%) occur at least once with both the  $[-n\partial k]$  and  $[-n\epsilon k]$  suffixes.

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Because we know the number tokens in the HNC (187.7mil) and each word's token frequency, we can estimate roughly how often speakers might hear vacillators. Assuming a speaker hears 10mil words per year (e.g., [Hart & Risley 2003](#)), over the course of an 80 year life only 28 vacillators would have been heard even 10 times. H&L's proposal that learners might learn vacillation rates type-by-type is untenable, undercut by sparsity ([Yang, 2013](#)).

Moreover, because type-by-type learning is only tenable for high frequency words, under a probabilistic grammar proposal, class-level probabilities should *leak* into the long-tail of the Zipfian distribution, producing vacillators from low token-frequency words. The learner, undercut by Zipf's law, would have no choice but to assign low frequency types class-level probabilities: words encountered rarely in the input are effectively wugs from the learner's point of view.

To show this visually, I plotted in Fig. 14, for each stem in the HNC, its fraction back value (in black). This is plotted against the probability of taking the back suffix predicted by MAXENT (orange) and the number of D2L $^{\pi}$  models where the stem was predicted to take a back suffix (blue). The figure sorts all rankings from 0 to 1 to give the overall shape of predicted vacillation. The actual distribution, and that predicted by D2L $^{\pi}$ , is a step-function: nearly all values are either 0 or 1, and only a handful of words are attested with both suffixes. In contrast, MAXENT incorrectly predicts an S-shaped curve, with an overly-gradient shift from the front-taking to back-taking stems. The Spearman's non-parametric rank correlation between the token-level fraction back values for each word in the HNC and those predicted by D2L $^{\pi}$  is  $\rho = 0.970$ , compared to  $\rho = 0.806$  between the HNC and MAXENT, which is a significant difference under a bootstrap significance test ( $p < 0.01$ ).

Under this close inspection, the appearance of some datives with both back and front variants is probably not reflective of individually-internalized token-level variation; rather their presence in the corpus more likely reflects subtle individual-level variation around the edges of the envelope of lexical variation, as D2L $^{\pi}$  predicts.

### 5.3. *The Argument from Opacification*

It has long been recognized that language acquisition plays a causal role in language change (e.g., [Halle 1962](#); [Kiparsky 1965](#); [Weinreich et al. 1968](#); [Lightfoot 1979](#); [Labov 1989](#); [Yang 2002](#); [Kodner 2020](#)). Thus, assuming continuity in the mechanisms of language change from the past to the present (the *Uniformitarian Principle*, [Lyell 1830](#); [Labov 1994](#); [Walkden 2019](#)), past acquisition outcomes serve as a testbed for proposals, like D2L, about learning mechanisms.

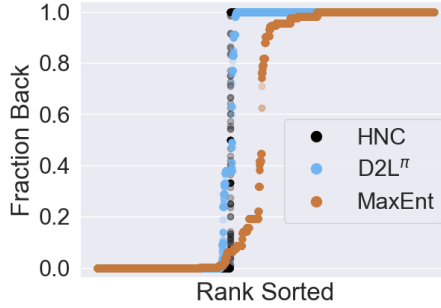


Figure 14: Predicted shape of vacillation distribution by MAXENT and  $D2L^\pi$ , compared to the HNC. The HNC distribution is more like the  $D2L^\pi$ -predicted step-like function than the more gradual S-shape curve predicted by MAXENT.

As Fig. 10 showed, of the neutral [i], [i:], [e:], and [ε],  $D2L^\pi$  predicts that [i] is *robustly transparent*, meaning learners consistently remove it from the tier. Similarly, [ε] is predicted to be *robustly opaque*, as learners consistently retain it on the tier. In contrast, [i:] and [e:] are comparatively *unstable*, as learners vary as to whether or not each is preserved on the tier as a sufficient predictor.

Because segments are only removed from the tier when they are *not* sufficient predictors, vowels that are not sufficient predictors in an internet-scale corpus may nevertheless sneak by as such for some learners, depending on the particular makeup of that learner’s lexicon. For instance, if a learner knows  $n_{e:} = 10$  words with [e:] as the tier-final segment,  $m_{e:} = 6$  of which take [-nɛk], [e:] will be retained as opaque (since  $4 \leq \theta_{10}$ ); the learner would need to come across more such words instead taking [-nɔk] to tip it over the threshold. Moreover, stems with all neutral vowels (e.g., N, NN stems) or all front vowels, neutral or not (e.g., FN, FNN stems) nearly categorically take the front suffix. It is only the BN(N) stems that work against a neutral vowel predicting [-nɛk]. This pressure towards opacity is compounded by the fact that the smaller the value of  $n$ , the *more tolerant* the threshold  $\frac{n}{\ln n}$  is (Yang, 2016, ch.3). For instance,  $n = 10$  allows  $4/10 = 40\%$  to be exceptions while  $n = 30$  allows only  $8/30 = 27\%$  exceptions. In short, the general diachronic pressure from D2L on neutral vowels on the porous periphery of the envelope of variation is *towards opacity*.

In artificial grammar learning settings, learners indeed show a learning pressure towards opacity over transparency. Finley (2015) found that adults learned the behavior of an opaque vowel but failed, under equivalent conditions, to learn the behavior of a transparent vowel. In other conditions, learners would succeed at learning transparent vowel behavior, but only if the amount of exposure to items that broke

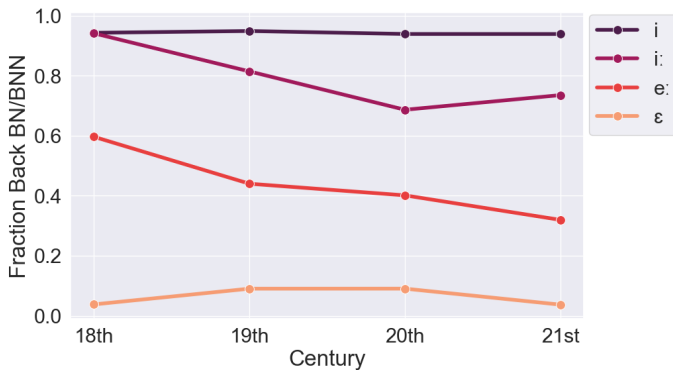


Figure 15: The vowels [i:] and [e:], which D2L<sup>π</sup> predicts to be diachronically susceptible to opacification over time, have been shifting in that direction.

the vowel’s predictive power was increased. In a later study, [Chen \(2024\)](#) found compatible results. Adults were trained on a harmony pattern with both an opaque and a transparent vowel. Depending on the exposure phase, the participants either failed to learn the harmony system altogether or appeared to treat both neutral vowels as opaque. [Belth \(2025\)](#) demonstrated that D2L accounts for this, for the above reason: transparent vowels can slip by as opaque.

It is possible to look into the past to see if [i:] and [e:], which D2L<sup>π</sup> predicts may opacify over time, indeed have been. The *Magyar történeti szövegtá*<sup>4</sup> (Hungarian Historical Text Library; henceforth MTSZ) comprises approximately 30 million tokens of Hungarian texts from the 18th-21st Centuries, covering several genres. The corpus is maintained by the Nyelvtudományi Kutatóközpont, who also maintains the HNC. From MTSZ, I extracted all datives (forms ending in -nak or -nek) for each century (18th, 19th, 20th, 21st). For each neutral vowel and century, I computed the fraction of BN/BNN stems that take the back suffix in that century.

The result is in Fig. 15. Since neutral vowels are front, a high fraction back value (towards 1.0) indicates a more *transparent* vowel and a value towards 0.0 indicates a more *opaque* vowel. A downward trend indicates opacification over time. Visually, the vowels [i:] and [e:] have been opacifying over time, precisely as predicted by D2L<sup>π</sup>.

To test statistically, I fit a Bayesian logistic regression model to the back/front suffix for each word, using century as the predictor. The 94% HDIs of each slope (the narrowest region containing 94% of the posterior probability over the estimates of each slope) are shown in Tab. 2. Only [i:] and [e:] show fully negative 94% HDIs while [i] and [ε] straddle 0, providing statistical support for the trends.

<sup>4</sup><https://nagyszotar.nyttud.hu/corpus.html>

Table 2: Statistical evaluation (94% HDIs) of the negativity of slopes in Fig. 15.

| Vowel | 94% HDI          | HDI Fully Negative |
|-------|------------------|--------------------|
| [i]   | [−0.159, 0.042]  | ✗                  |
| [i:]  | [−0.782, −0.059] | ✓                  |
| [e:]  | [−0.416, −0.225] | ✓                  |
| [ɛ]   | [−0.169, 0.158]  | ✗                  |

While I am not aware of any prior reports of the diachronic opacification of [i:] and [e:], the observation is specific enough that it is unlikely that, absent a theory like  $D2L^\pi$  that predicts it, anyone would have thought to investigate. This opacification is not incompatible with frequency-matching: if lexical frequencies shift over time, frequency-matchers could track the shifting distribution. However, it is not possible for frequency matching in itself to predict the particular shifts—something additional would be required to cause it. In contrast,  $D2L^\pi$  provides a mechanistic explanation for the specific shift. The fact that the high vowels [i] and [i:] behave importantly differently here, as in § 4.4, provides further evidence that the categories of interest in H&L and HZSL are indeed too coarse, analogous to the example of shapes in § 2.3.

6. Discussion

In this paper, I have pursued two intertwined contributions, one critical and one constructive. I began with a critical analysis of the Frequency Matching Hypothesis in the context of lexical variation, which asserts that a core component of phonological learning is veridical matching of lexical frequencies by means of a probabilistic grammar. The criticism focused on H&L and HZSL’s influential studies. It first contended that, counterintuitively, probabilistic behavior does not imply a probabilistic mechanism (§ 2.1), so even if it is fair to characterize Hungarian speakers’ behavior as *frequency matching* there is no basis for concluding that they must be internalizing a probabilistic grammar. Secondly, H&L and HZSL’s wug test results are group-level aggregates. Any appearance of frequency matching at the group level cannot be used as the basis for a claim about individual-level frequency matching (even descriptively), because averaged, group-level measures do not in general approximate individual measures. H&L’s attempt to argue that in their case the group is reflective of the individuals falls short (§ 2.2). Thirdly, measuring frequency matching behavior—whether group or individual-level—requires an assumption about the categories underlying the variation—i.e., what counts as being “the same type” with respect to the pattern. Faulty

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assumptions about the underlying categories can lead categorical behavior to look like frequency matching, and prior studies of Hungarian vowel harmony may have made such an error (§ 2.3, § 4.4, § 5.3). Even if we take H&L’s and HZSL’s results at face value, there is no evidence of individual-level frequency matching. The wug tests involved insufficient repeated measures to distinguish between categorical behavior and frequency matching (§ 2.4).

I suspect other works reported as evidence of frequency matching may be susceptible to similar criticisms. For example, Ernestus & Baayen (2003)’s study of Dutch voicing alternations and Zuraw (2000, 2010)’s study of Tagalog nasal substitution—both often cited as evidence of frequency matching—are also based on group-level aggregates and tacit assumptions about the relevant categories. Even if frequency matching is a fair characterization of group behavior in any such studies, none implies a central probabilistic mechanism underlying phonological learning any more than rats probability matching a T-maze implies a probabilistic mechanism (§ 2.1). I conjecture that frequency matching is less law-like than reports suggest.

While I disagree that there is substantial evidence for the Frequency Matching Hypothesis, I agree with much of this line of work—including H&L and HZSL—that productive structural conditioning of lexical variation enters into the grammar rather than variation being idiosyncratically marked in the lexicon (e.g., this is a central point of Zuraw 2010 and Gouskova & Becker 2013). In fact, this is the starting point for the constructive part of the study—an independently-motivated learning theory for the structural conditioning of lexical variation, which lends itself to a *variationist* theory of frequency matching, where Hungarian wug test results are indicative of variation across speakers rather than individuals each internalizing a probabilistic grammar. The learning proposal is based on D2L (Belth, 2026), which uses as its central mechanism the well-established ability to track adjacent dependencies. This core generalization mechanism automatically yields tier-like representations, where tier-adjacent predictors of the [-nɔk]/[-nɛk] alternation partition the input into equivalence classes constituting the learned structural conditioning of lexical variation.

D2L is motivated on psychological grounds independent of Hungarian, and has been applied to learning a range of phonological alternations (§ 5.1). Through the interpretability of the learned structural conditioning, it is possible to investigate the amount and nature of variation across learners (§ 4.4).  $D2L^\pi$  predicts that the few words attested with both suffixes in corpus data are indicative of variation across the speakers contributing to the corpus, rather than genuine token-level variation. Careful

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inspection of the corpus statistics supports this position (§ 5.2). Investigation of variation across D2L<sup>π</sup> learners also predicts that the neutral vowels [i:] and [e:], but not [i] and [ε], are subject to variation across learners that introduces a pressure towards *opacification*, where the vowels become greater predictors of the front suffix over time. Analysis of a historical corpus does show evidence for opacification over time (§ 5.3). These serve as positive arguments for this variationist, learning-based proposal.

There is a certain paradox that emerges. One of the most celebrated features of statistical models is their ability to capture variation (Coetzee & Pater, 2011; Pater, 2019). Yet it is precisely a probabilistic grammar that underlies H&L and HZSL's non-variationist position that group averages are the primary object of study—that “the level of variation is not between individuals, but within the individual” (H&L, p. 74). Instead it is D2L<sup>π</sup>, in which each model behaves categorically, that embodies a variationist theory of Hungarian vowel harmony. I conjecture that the reason for this paradox might be that the placement of probabilities into a grammar, and the privileging of statistical learning, flattens variation in corpus and experimental data. But datasets are typically the product of populations of speakers, not individuals, and observed variation is multi-leveled and heterogeneous in source and structure. It is a central task of the linguist to decompose the web of structured heterogeneity underlying language, as it is for language learners (Weinreich *et al.*, 1968). It is my hope that a focus on the central learning questions of how the structure underlying heterogeneity is discovered, such as how learners discover the structural conditioning of lexical variation, will provide a stronger footing for a theory of variation than a focus on endowing models with probabilities, which side steps the fundamental questions about underlying structure.

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